Tousik Samui

The Institute of Mathematical Sciences (IMSc),
IV Cross Road, CIT Campus, Taramani, Chennai 600 113, Tamil Nadu, India.
tousiks@imsc.res.in • tousiksamui@gmail.com
Webpages: tousiksamui.github.io • www.iiserkol.ac.in/∼tousik.pdf/ • www.imsc.res.in/tousik_samui

Current Position

Post Doctoral Fellow June 2024 – Present
The Institute of Mathematical Sciences (IMSc), Chennai, India.

Previous Positions

Post-doctoral Fellow November 2021 – May 2024
Indian Institute of Science Education and Research Kolkata (IISER-K), Mohanpur, India.

Post-Doctoral Fellow September 2018 – November 2021
Harish-Chandra Research Institute (HRI), Prayagraj, India.

Academic Background

Ph.D. in Physics 2019
Tata Institute of Fundamental Research (TIFR), Mumbai, India.

M.Sc. in Physics 2012
Indian Institute of Technology Kanpur (IITK), Kanpur, India.

B.Sc. (Honours) in Physics 2010
University of Calcutta (CU), Kolkata, India.

Research Interests

Physics Beyond the Standard Model, Electroweak Interactions, Collider Physics, Jet and Jet Substructure at Colliders, Dark Matter.

Special Achievements

Notable Ranks in National Exams:

- *All India Rank* **3**, National Eligibility Test (NET) in Physical Sciences. June 2013
- *All India Rank* **15**, Graduate Aptitude Test in Engineering (GATE) in Mathematics. 2013
- *All India Rank* **43**, National Eligibility Test (NET) in Mathematical Sciences. December 2012
- *All India Rank* **16**, Joint Entrance Screening Test (JEST) in Physics. 2011
- *All India Rank* **23**, Graduate Aptitude Test in Engineering (GATE) in Physics. 2011
- *All India Rank* **5**, Joint Admission test for Masters (JAM) in Physics. 2010

Fellowships:

- *Dr. Shyama Prasad Mukherjee Fellowship in Physical Sciences*, awarded by the Council of Scientific and Industrial Research, Government of India. 2013
- *Junior Research Fellowship in Mathematical Sciences*, awarded by the Council of Scientific and Industrial Research, Government of India. 2013

- *Inspire Scholarship*, awarded by the Department of Science and Technology, Government of India. 2007 – 2012

Awards:

- *DPS day 2023 Best Poster*, awarded by the Department of Physical Sciences, IISER-K, India. March 2023
- *Academic Excellence in the M.Sc. (2-year) Programme in Physics*, awarded by the Indian Institute of Technology Kanpur, Kanpur, India. 2011

Computer Skills

Computer languages: Fortran, C, C++, Python, PHP, HTML.

Packages in hep-ph: CalcHEP, FastJet, FeynRules, Delphes, MadGraph5, MicrOmegas, Pythia8, Root, SARAH, Spheno, SSP.

Research Skills

- BSM model building and phenomenology.
- Studies of collider signatures for SM and BSM searches.
- Jet and jet substructure based studies.
- Development of modules/algorithm for collider studies.
- Machine learning and handling of HEP packages.

Academic Visits

- Indian Institute of Science Education and Research Mohali, Manauli, India. February 2026.
- Indian Institute of Science Education and Research Kolkata, Mohanpur, India. August – September 2025
- CERN Theoretical Physics Department, Geneva, Switzerland. June – July 2025
- Indian Institute of Science Education and Research Pune, Pune, India. February 2024
- Indian Institute of Science, Bengaluru, India. September 2023
- Harish-Chandra Research Institute, Prayagraj, India. March 2022
- Laboratoire de Physique Théorique d'Orsay, Orsay, France. June 2017
- Scuola Normale Superiore, Pisa, Italy. June 2017
- Laboratoire de Physique Subatomique and Cosmologie, Grenoble, France. June 2017
- Harish-Chandra Research Institute, Prayagraj, India. March – May 2018

Teaching Activities

- Online lectures on **Data handling using ROOT** at School and Workshop on Statistical Methods and Deep Machine Learning in High Energy Physics and Astrophysics (Pre-school, online), IISER Kolkata, Mohanpur, India. Link: https://www.iiserkol.ac.in/~ml4hep, May 2025.
- Online lectures (together with Dr. Ratul Mahanta) on **Introduction and Quantization of Fields** at PWF Bangladesh: School on Quantum Field Theory, ICTP, Trieste, Italy. Link: https://indico.ictp.it/event/10720 and https://ahmedrakinkamal.com/physics-for-bangladesh, Aug 25, 2024 – Oct 1, 2024.

- Lecture series on **Linear Representations of Groups**, IISER Kolkata, Mohanpur, India. (September 2022 – October 2022).
- Teaching Assistantship: **Mechanics I**, IISER Kolkata, Mohanpur, India. Autumn 2022
- Teaching Assistantship: **Electrodynamics II**, TIFR, Mumbai, India. Autumn 2015
- Teaching Assistantship: **Quantum Field Theory I**, TIFR, Mumbai, India. Spring 2015
- Teaching Assistantship: **Classical Mechanics**, TIFR, Mumbai, India. Autumn 2013

Additional Activities

- Secretary of Machine-learning Optimized Design of Experiments (MODE) Collaboration. December 2023 – November 2025
- Organizing member of *Weekly High Energy Physics Journal club Discussion* at IISER Kolkata, Mohanpur, India. August 2023 – May 2024
- System Administrator and Maintenance Coordinator of the department's computing cluster Kepler, IISER Kolkata, Mohanpur, India. April 2023 – May 2024
  https://www.iiserkol.ac.in/~dps.hpc
- Active member of Indian Physics Club (a discussion group consisting of some young researchers in India). August 2020 – September 2021
  sites.google.com/view/indianphysicsclub/home
- Designed and maintained a webpage which shows the live status of running jobs in HRI-HPC cluster, HRI, Prayagraj, India. January 2020 – November 2021
  http://www.hri.res.in/cluster/clusterStatus/cluster.php
- Coordinator and Volunteer Teacher in Harsha School (a welfare activity managed by students and post-docs of HRI), HRI, Prayagraj, India. April 2019 – November 2021
- Organised weekly HEP Journal Club at HRI, Prayagraj, India. October 2018 – June 2020
- Canteen Secretary of TIFR Students' Society (Student body of TIFR), TIFR, Mumbai, India. April 2014 – May 2015

Online Profiles

Webpages: tousiksamui.github.io; www.iiserkol.ac.in/~tousik.pdf; www.imsc.res.in/tousik_samui
InspireHEP: inspirehep.net/authors/1413674
arXiv: arxiv.org/a/samui_t_1.html
ORCID: 0000-0002-1485-6155
Google Scholar: scholar.google.com/citations?user=7_l0Ty4AAAAJ
Scopus: www.scopus.com/authid/detail.uri?authorId=57194699349

References are available on request.

Publications and Preprints

(In High Energy Physics, authors are listed alphabetically, and there is no traditional first author. Asterisk symbols (*) represent the corresponding authors.)

Publications

1. Nivedita Ghosh *, Santosh Kumar Rai, **Tousik Samui**, Agnivo Sarkar 2026
Investigating the leptonic couplings of doubly charged scalars at the muon collider
Physical Review D **113** (2026) 055048 • DOI: 10.1103/jbq9-jy42

2. Anupam Ghosh, Soumyadip Ghosh, Soureek Mitra, **Tousik Samui** *, Ritesh K. Singh 2026
Improving Sensitivity of Vectorlike Top Partner Searches with Jet Substructure
Physical Review D **113** (2026) 035034 • DOI: 10.1103/dhs6-gcn9

3. Anjan Kumar Barik, Atri Dey, **Tousik Samui** * 2026
Search for Lepton Flavor Violating Signals at the Future Electron-Proton Colliders
Physics Letters B **872** (2026) 140048 • DOI: 10.1016/j.physletb.2025.140048

4. Anupam Ghosh *, Partha Konar, **Tousik Samui**, Ritesh K. Singh 2025
Jet substructure probe on scalar leptoquark models via top polarization
Journal of High Energy Physics **07** (2025) 145 • DOI: 10.1007/JHEP07(2025)145

5. Songshaptak De, Atri Dey, **Tousik Samui** * 2025
Jet Substructure Analysis for Distinguishing Left- and Right-Handed Couplings of Heavy Neutrino in W' Decay at the HL-LHC
Physical Review D **111** (2025) 075017 • DOI: 10.1103/PhysRevD.111.075017

6. Waleed Abdallah, Anjan Kumar Barik *, Santosh Kumar Rai, **Tousik Samui** 2024
Heavy Neutrino as Dark Matter in a Neutrinophilic $U(1)$ Model
The European Physical Journal C **84** (2024) 1087 • DOI: 10.1140/epjc/s10052-024-13407-z

7. Biswarup Mukhopadhyaya, Sirshendu Samanta *, **Tousik Samui** *, Ritesh K. Singh 2024
Improving probes of hAA coupling in the Type-X two Higgs doublet model scenario: The crucial role of τ-jet charge identification
Physics Letters B **856** (2024) 138949 • DOI: 10.1016/j.physletb.2024.138949

8. Nivedita Ghosh *, Santosh Kumar Rai, **Tousik Samui** 2024
Search For a Leptoquark and Vector-like Lepton in a Muon Collider
Nuclear Physics B **1004** (2024) 116564 • DOI: 10.1016/j.nuclphysb.2024.116564

9. Atri Dey, **Tousik Samui** * 2023
Polarization study of a boosted W boson decaying hadronically at the LHC with jet substructures and a multivariate analysis
The European Physical Journal C **83** (2023) 1002 • DOI: 10.1140/epjc/s10052-023-12118-1

10. Biswarup Mukhopadhyaya, Sirshendu Samanta, **Tousik Samui** *, Ritesh K. Singh 2023
Novel signals for the Type-X two Higgs doublet scenario at the Large Hadron Collider
Physical Review D **108** (2023) 075004 • DOI: 10.1103/PhysRevD.108.075004

11. Biswarup Mukhopadhyaya, **Tousik Samui** *, Ritesh K. Singh 2023
Dynamic Radius Jet Clustering Algorithm
Journal of High Energy Physics **04** (2023) 019 • DOI: 10.1007/JHEP04(2023)019

12. Nivedita Ghosh *, Santosh Kumar Rai, **Tousik Samui** 2023
Collider signatures of a scalar leptoquark and vectorlike lepton in light of muon anomaly
Physical Review D **107** (2023) 035028 • DOI: 10.1103/PhysRevD.107.035028

13. Waleed Abdallah, Anjan Kumar Barik, Santosh Kumar Rai *, **Tousik Samui** 2023
Emergent new symmetry from the Higgs shadow
Physical Review D **107** (2023) 015026 • DOI: 10.1103/PhysRevD.107.015026

14. Waleed Abdallah, Anjan Kumar Barik, Santosh Kumar Rai, **Tousik Samui** * 2021
Search for a light Z' at LHC in a neutrinophilic U(1) model
Physical Review D **104** (2021) 095031 • DOI: 10.1103/PhysRevD.104.095031

15. **Tousik Samui**, Nilakash Sorokhaibam * 2021
Thermalization in different phases of charged SYK model
Journal of High Energy Physics **04** (2021) 157 • DOI: 10.1007/JHEP04(2021)157

16. Aruna Kumar Nayak, Santosh Kumar Rai, **Tousik Samui** * 2021
Improving Heavy Dijet Resonance Searches Using Jet Substructure at the LHC
The European Physical Journal C **81** (2021) 130 • DOI: 10.1140/epjc/s10052-021-08856-9

17. Suman Chatterjee, Anirban Das *, **Tousik Samui**, Manibrata Sen 2019
Mixed WIMP-axion dark matter
Physical Review D **100** (2019) 115050 • DOI: 10.1103/PhysRevD.100.115050

18. Amit Chakraborty, Ushoshi Maitra, Sreerup Raychaudhuri, **Tousik Samui** * 2017
Mixed Higgs–radion states at the LHC – a detailed study
Nuclear Physics B **922** (2017) 41 • DOI: 10.1016/j.nuclphysb.2017.06.006

Proceedings

1. Waleed Abdallah, Anjan Kumar Barik *, Santosh Kumar Rai, **Tousik Samui** 12–16 Dec. 2022
Emergent New Symmetry from the Higgs Shadow
Contribution to: 25th DAE-BRNS High Energy Physics Symposium 2022, IISER Mohali, Punjab, India.
Springer Proceedings in Physics **304** (2024) 746 • DOI: 10.1007/978-981-97-0289-3_182

2. J.R. Andersen, J. Bellm, J. Bendavid, N. Berger, D. Bhatia, **T. Samui** et al. 5–23 Jun. 2017
Les Houches 2017: Physics at TeV-2mmColliders Standard Model Working Group Report
Contribution to: Physics at TeV Colliders 2017, Les Houches, France.
Proceedings of: PhysTeV 2017 • e-Print: arxiv:1803.07977 [hep-ph]

Preprints

1. Songshaptak De, **Tousik Samui**, Ritesh K. Singh
Generalised Dynamic Radius Jets for Robust Collider Analyses

e-Print:arXiv:2609.29862

2. Pankaj Agrawal, Biswajit Das, **Tousik Samui**
Associated Higgs production in lepton-photon collisions at FCC-ee
e-Print:arXiv:2603.27822

3. Songshaptak De, Atri Dey, **Tousik Samui**
Unfolding quantum entanglement from $h \to ZZ^* \to jj\ell\ell$ at a muon collider
e-Print: arXiv:2512.09121 [hep-ph]

4. Debjyoti Bardhan, Disha Bhatia, Amit Chakraborty, Ushoshi Maitra, Sreerup Raychaudhuri, **Tousik Samui**
Radion Candidate for the LHC Diphoton Resonance
e-Print: arxiv:1512.06674 [hep-ph]

Research software/Technical contributions

1. **Tousik Samui** *
DYNAMICR: Official plugin for Dynamic Radius Jet Clustering Algorithm (JHEP **04** (2023) 019) included in FastJet Contrib (FJContrib v1.104, released July 2026).

Conferences and Workshops Attended

1. **Evolving Beyond the Standard Model**, European Organization for Nuclear Research (CERN), Geneva, Switzerland. June – July 2025
2. **Physics at TeV Colliders and Beyond the Standard Model 2025 (SM session)**, École de physique des Houches, Les Houches, France. June 2025
3. **School and Workshop on Statistical Methods and Deep Machine Learning in High Energy Physics and Astrophysics (Pre-school, online)**, Indian Institute of Science Education and Research Kolkata, Mohanpur, India. May 2025
4. **Journeys in Particle Physics - Rohini Godbole Memorial Conference**, Indian Institute of Science, Bengaluru, India. March 2025
5. **IMSc Spring School on High Energy Physics**, The Institute of Mathematical Sciences, Chennai, India. February-March 2025
6. **Conference on Cosmology, Astrophysics, and Particle Physics 2025**, SRM Institute of Science and Technology, Chennai, India. January-February 2025
7. **Current Trends in Particle and Astroparticle Physics 2025**, Indian Association for the Cultivation of Science, Kolkata, India. January 2025
8. **XXVI DAE-BRNS High Energy Physics Symposium - 2024**, Banaras Hindu University, Varanasi, India. December 2024
9. **Trends in Astroparticle and Particle Physics (TAPP) 2024**, The Institute of Mathematical Sciences, Chennai, India. September 2024
10. **Frontiers in Particle Physics (FPP) 2024**, Indian Institute of Science, Bengaluru, India. August 2024
11. **Frontier Topic in Collider Physics (FTCP 2024)**, Indian Institute of Science Education and Research Pune, Pune, India. February 2024
12. **Workshop on High Energy Physics Phenomenology XVII**, Indian Institute of Technology Gandhinagar, Gandhinagar, India. January 2024
13. **Phoenix 2023**, Indian Institute of Technology Hyderabad, Hyderabad, India. December 2023
14. **Statistical Methods and Machine Learning in High Energy Physics (ML4HEP)**, International Centre for Theoretical Sciences, Bengaluru, India. August-September 2023
15. **PP65: Physics Trends @ IISER Kolkata**, Indian Institute of Science Education and Research Kolkata, Mohanpur, India. June 2023
16. **International Meeting on High Energy Physics II (IMHEP-II)**, Institute of Physics, Bhubaneswar, India. February 2023
17. **HSF-India Research Software Networks in Physics**, S. N. Bose National Centre for Basic Sciences, Kolkata, India. January 2023
18. **32nd Meeting of Indian Association for General Relativity and Gravitation**, Indian Institute of Science Education and Research Kolkata, Mohanpur, India. December 2022
19. **November Meeting at IISER-K**, Indian Institute of Science Education and Research Kolkata, Mohanpur, India. November 2022

20. **Anomalies 2021** (online), Indian Institute of Technology Hyderabad, Hyderabad, India. September 2021
21. **Anomalies 2020** (online), Indian Institute of Technology Hyderabad, Hyderabad, India. September 2020
22. **Madgraph School 2019**, The Institute of Mathematical Sciences, Chennai, India. November 2019
23. **Academic Program – HEP activity weeks**, Harish-Chandra Research Institute, Prayagraj, India. March 2019
24. **International Meeting on High Energy Physics (IMHEP)**, Institute of Physics, Bhubaneswar, India. January 2019
25. **2nd Workshop of the Indo-French Network in High Energy Physics**, Indian Institute of Science Education and Research Pune, Pune, India. February 2018
26. **Blueprints Beyond the Standard Model**, Tata Institute of Fundamental Research, Mumbai, India. January 2018
27. **15th Workshop on High Energy Physics Phenomenology**, Indian Institute of Science Education and Research Bhopal, Bhopal, India. December 2017
28. **25th International Conference on Supersymmetry and Unification of Fundamental Interactions (SUSY 17)**, Tata Institute of Fundamental Research, Mumbai, India. December 2017
29. **Mumbai-Pune Collider Meet**, Indian Institute of Technology Bombay, Mumbai, India. October 2017
30. **Physics at TeV colliders**, Les Houches, France. June 2017
31. **Pune-Mumbai Collider Meet**, Indian Institute of Science Education and Research, Pune, India. December 2016
32. **Looking for BSM Physics**, Indian Institute of Science, Bengaluru, India. December 2016
33. **9th International Workshop on the CKM Unitarity Triangle**, Tata Institute of Fundamental Research, Mumbai, India. November 2016
34. **Indo-French LIA THEP and CEFIPRA INFRE-HEPNET Kick-Off Meeting**, Indian Institute of Science, Bangalore, India. May 2016
35. **Workshop on LHC and Dark Matter**, Indian Association for the Cultivation of Science, Kolkata, India. February 2015
36. **XXIX SERC Main School on Theoretical High Energy Physics**, Birla Institute of Technology and Science, Goa, India. December 2014 – January 2015
37. **XXI DAE-BRNS High Energy Physics Symposium 2014**, Indian Institute of Technology Guwahati, Guwahati, India. December 2014.
38. **Sangam@HRI 2014**, Harish-Chandra Research Institute, Prayagraj, India. March 2014
39. **Lecture Workshop on High Energy Physics**, Indian Institute of Technology Bombay, Mumbai, India. January 2013
40. **TeV Particle Astrophysics 2012**, Tata Institute of Fundamental Research, Mumbai, India. December 2012

Invited and Contributed Talks

1. **Enhancing Collider Investigations with Dynamic Radius Jets**, talk at CMS B2G Group Meeting (online). April 28, 2026
2. **Jets and Jet Substructure Adventures at High Energy Colliders**, seminar at Indian Institute of Science Education and Research Mohali, Manauli, India. February 26, 2026
3. **Prospects of Dynamic Radius Jet Algorithm at the LHC**, led discussion at Physics at TeV Colliders and Beyond the Standard Model 2025 (SM session), Les Houches, France. June 20, 2025
4. **Jets and Jet Substructure: Advanced Tools for Collider Searches**, seminar at The Institute of Mathematical Sciences. Chennai, India. March 20, 2025
5. **Dynamic Radius Jets at the LHC**, contributed talk at Conference on Cosmology, Astrophysics, and Particle Physics 2025, Chennai, India. February 1, 2025
6. **Prospects of Dynamic Radius Jet Clustering Algorithm at the LHC**, contributed talk at XXVI DAE-BRNS High Energy Physics Symposium - 2024, BHU, Varanasi, India. December 22, 2024
7. **Prospects of Dynamic Radius Jet Clustering Algorithm at the LHC**, contributed talk at Trends in Astroparticle and Particle Physics (TAPP) 2024, IMSc, Chennai, India. September 25, 2024
8. **Dynamic Radius Jet Clustering Algorithm**, contributed talk at Frontiers in Particle Physics (FPP) 2024, IISc, Bengaluru, India. August 9, 2024
9. **Polarization study of a boosted W boson decaying hadronically at the LHC with jet substructures and a multivariate analysis**, talk at ATLAS L2IT Physics Analysis Meeting (online). June 6, 2024
10. **An overview of DPS High Performance Computing Facility**, contributed talk at DPS Days 2024, IISER Kolkata, Mohanpur, India. March 9, 2024
11. **Exploring the Type-X Two Higgs Doublet Model at the LHC: the Crucial Role of tau-jet Charge Identification**, contributed talk at DPS Days 2024, IISER Kolkata, Mohanpur, India. March 8, 2024
12. **Dynamic Radius Jet Clustering Algorithm**, seminar at Indian Institute of Science Education and Research Pune, Pune, India. February 21, 2024
13. **Jets and Beyond the Standard Model Adventures at Colliders**, contributed talk at DPS Research Meet 2024, IISER Kolkata, Mohanpur, India. January 20, 2024
14. **Managing Colour Flow in ML**, led discussion at Workshop on High Energy Physics Phenomenology XVII, IIT Gandhinagar, Gandhinagar, India. January 10, 2024
15. **Prospects of Dynamic Radius Jet Clustering Algorithm**, led discussion at Workshop on High Energy Physics Phenomenology XVII, IIT Gandhinagar, Gandhinagar, India. January 5, 2024
16. **Dynamic Radius Jet Clustering Algorithm**, contributed talk at Phoenix 2023, IIT Hyderabad, Hyderabad, India. December 19, 2023
17. **Dynamic Radius Jet Clustering Algorithm**, seminar at Indian Institute of Science, Bengaluru, India. September 13, 2023

18. **Search for Rare Process using Self Organising Maps at the LHC**, Work Progress Report at Statistical Methods and Machine Learning in High Energy Physics (ML4HEP), ICTS, Bengaluru, India. September 8, 2023
19. **Lepton Flavour Violating Signals at the Future Electron-Proton Colliders**, contributed talk at PP65: Physics Trends @ IISER Kolkata, IISER Kolkata, Mohanpur, India. June 28, 2023
20. **Dynamic Radius Jet Clustering Algorithm**, contributed talk at International Meeting on High Energy Physics II (IMHEP-II), IOP, Bhubaneswar, India. February 22, 2023
21. **Jet Substructure-based Studies at the LHC**, led discussion at November Meeting at IISER-K, IISER Kolkata, Mohanpur, India. November 18, 2022
22. **Jetography: Revolutionizing Collider Era**, seminar at Gluon – The Physics Journal Club, IISER Kolkata, Mohanpur, India. June 17, 2022
23. **Polarization Study of Boosted, Hadronic W using Jet Substructure at the LHC**, seminar at Harish-Chandra Research Institute, Prayagraj, India. March 24, 2022
24. **Jet Substructure and Multivariate Analysis Aid in Polarization Study of Boosted, Hadronic W at the LHC**, (online) contributed talk at Anomalies 2021, IIT Hyderabad, Hyderabad, India. November 11, 2021
25. **Impact of Jet Substructure Technique in Heavy Resonance Searches at the LHC**, seminar at Harish-Chandra Research Institute, Prayagraj, India. September 22, 2020
26. **Improving Heavy Dijet Resonance Searches Using Jet Substructure at the LHC**, (online) contributed talk at Anomalies 2020, IIT Hyderabad, Hyderabad, India. September 12, 2020
27. **Probing Heavy Resonances Using Jet Substructure at the LHC**, seminar at Harish-Chandra Research Institute, Prayagraj, India. September 11, 2019
28. **Missing Energy Signal in $t\bar{t}$ to Probe Large Extra Dimensions at Collider**, contributed talk at Academic Program – HEP activity weeks, HRI, Prayagraj, India. March 12, 2019
29. **Jet Substructure at the LHC**, seminar at Harish-Chandra Research Institute, Prayagraj, India. April 4, 2018
30. **Mixed Higgs-Radion States at the LHC**, contributed talk at DTP Interaction Meeting 2018, TIFR, Mumbai, India. February 9, 2018
31. **Mixed Higgs-Radion States at the LHC**, contributed talk at Blueprints Beyond the Standard Model, TIFR, Mumbai, India. January 8, 2018
32. **Mixed Higgs-Radion States at the LHC**, led discussion at 15th Workshop on High Energy Physics Phenomenology, IISER Bhopal, Bhopal, India. December 22, 2017
33. **Missing Energy Signals in $t\bar{t}$ Production as a Probe of Large Extra Dimensions at an e^+e^- Collider**, contributed talk at 25th International Conference on Supersymmetry and Unification of Fundamental Interactions (SUSY 17), TIFR, Mumbai, India. December 11, 2017
34. **Missing Energy Signal in $t\bar{t}$ to Probe Large Extra Dimensions at e^+e^- Colliders**, seminar at Tata Institute of Fundamental Research, Mumbai, India. November 2, 2017
35. **Missing Energy Signal in $t\bar{t}$ to Probe Large Extra Dimensions at ILC**, contributed talk at Mumbai-Pune Collider Meet, IIT Bombay, Mumbai, India. October 15, 2017
36. **Mixed Higgs-Radion States at the LHC**, seminar at Tata Institute of Fundamental Research, Mumbai, India. October 12, 2017

37. **Mixed Higgs-Radion States at the LHC**, seminar at Laboratoire de Physique Théorique d'Orsay, Orsay, France. June 27, 2017
38. **Mixed Higgs-Radion States at the LHC**, seminar at Scuola Normale Superiore, Pisa, Italy. June 21, 2017
39. **Mixed Higgs-Radion States at the LHC**, seminar at Laboratoire de Physique Subatomique et de Cosmologie, Grenoble, France. June 15, 2017
40. **Missing energy in $t\bar{t}$ production at e^+e^- Collider as a probe of Large Extra Dimensions**, contributed talk at XXI DAE-BRNS High Energy Physics Symposium, IIT Guwahati, India. December 9, 2014